



Friday Mar-21-2025

**Flowserve Facility:**

Flowserve Nordstrom / Valbart Valves  
1511 Jefferson St.  
Sulphur Springs, TX 75482

cmtrs@flowserve.com

**To:**

***CERTIFICATE OF CONFORMANCE***

| SERIAL #  | HEAT # | PO #    | Description | Class | End Connection |
|-----------|--------|---------|-------------|-------|----------------|
| 61-689550 | 4026   | 1170871 | 4" 2345     | 900   | FLG            |
| 61-689551 | 4027   | 1170871 | 4" 2345     | 900   | FLG            |
| 61-689552 | 4012   | 1170729 | 4" 2345     | 900   | FLG            |
| 61-689553 | 0299B  | 1170729 | 4" 2345     | 900   | FLG            |
| 61-689554 | 3977   | 1170729 | 4" 2345     | 900   | FLG            |

We hereby certify that all the subject valves/parts/sealant have been manufactured in conformance with the applicable requirements of Flowserve Sulphur Springs Operation (FSSO) specifications and that the valve/parts/sealant materials for component parts meet the applicable requirements of FSSO drawings and ASTM specifications, ASME B16.34 (If Applicable), API Q1 10th edition, and API 6D 25th edition.(if applicable to the API monogram). Flowserve Dynamic Balanced Valves are provided with a Pressure balance feature unless specifically indentified in the Valve description. All valves/parts/sealant meet all the requirements of the customer purchase order.

Documentation Technician  
Justin McKee

|                                                                                                       |                                                                                                                                     |  |                                                              |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--|--------------------------------------------------------------|
| <br><b>CASTLEGATE</b> | <b>CASTLEGATE INDUSTRIAL HOLDINGS PRIVATE LIMITED</b>                                                                               |  | <b>ISO 9001 : 2015</b><br><b>CERTIFIED</b><br><b>COMPANY</b> |
|                                                                                                       | S.F.No.387/1, Devanampalayam(P.O), Pollachi(TK), Coimbatore, 642 120<br>Phone :7397161500 04259264166, Mail :info@castlegatellc.com |  |                                                              |
|                                                                                                       | <b>INSPECTION CERTIFICATE</b><br><b>According to EN10204 - 3.1</b>                                                                  |  |                                                              |

|                                                               |                                                                                           |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Customer</b> : FLOWSERVE INDIA CONTROLS PVT LTD<br>CHENNAI | <b>T.C. No</b> : FLS/4026 - 001<br><b>T.C.Date</b> : 13/05/2024                           |
| <b>PO.No</b> : 10116580 Dt. 20/02/2024                        | <b>Heat No</b> : 4026<br><b>Pour Date</b> : 13/04/2024                                    |
| <b>Specification</b> : ASTM A216 GR.WCC (UNS J02503) - 2021   | CASTING PROCESSED BY INDUCTION MELTING FURNACE<br><br><b>FOUNDRIY IDENTIFICATION : CG</b> |

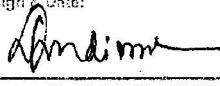
| CHEMICAL COMPOSITION % |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
|------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Element                | C     | Mn    | Si    | P     | S     | Cr    | Ni    | Mo    | Cu    | V     | CE    | RE    | AL    | N     |
| Minimum                | ---   | ---   | ---   | ---   | ---   | ---   | ---   | ---   | ---   | ---   | ---   | ---   | ---   | ---   |
| Maximum                | 0.230 | 1.200 | 0.600 | 0.025 | 0.020 | 0.500 | 0.500 | 0.200 | 0.300 | 0.030 | 0.430 | 1.000 | 0.080 | 0.015 |
| Actual                 | 0.189 | 0.916 | 0.408 | 0.023 | 0.009 | 0.055 | 0.039 | 0.003 | 0.007 | 0.007 | 0.358 | 0.111 | 0.052 | 0.006 |

|                         |                                                                                                                                |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>HEAT TREATMENT :</b> | NORMALISING: Temperature raised to 920°C, soaked for 4 Hrs and then atmospheric air cooled.(Cycle No:C0018/24-25Dt.16/04/2024) |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------|

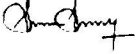
| MECHANICAL PROPERTIES |                |        |                  |        |             |             |          |           |                             |     |     |     |
|-----------------------|----------------|--------|------------------|--------|-------------|-------------|----------|-----------|-----------------------------|-----|-----|-----|
| Requirements          | Yield Strength |        | Tensile Strength |        | %Elongation | % Reduction | Hardness | Bend Test | Impact Test                 |     |     |     |
|                       | kgf/mm²        | Mpa    | kgf/mm²          | Mpa    | GL :50 mm   | In Area     | HBW      |           | CVN 2mm at MIN AVG 27 JULES |     |     |     |
|                       | 0.2 % Rp       |        | -                |        | -           | -           | -        | -         |                             |     |     |     |
| Minimum               | 28.03          | 275    | 49.44            | 485    | 22          | 35          | ---      | ---       |                             |     |     |     |
| Maximum               | ---            | ---    | 66.77            | 655    | ---         | ---         | 235      | ---       | 1                           | 2   | 3   | Avg |
| Actual                | 32.73          | 321.04 | 55.87            | 548.06 | 30.4        | 62.98       | 156      | ---       | ---                         | --- | --- | --- |

| ITEM DESCRIPTION |                     |                 |            |          |
|------------------|---------------------|-----------------|------------|----------|
| SL.No.           | Description         | Drawing No.     | Part No    | Quantity |
| 1                | BODY CSTG 4" CL 900 | 00487876 REV 21 | N487876WCC | 4        |

|                                                                  |
|------------------------------------------------------------------|
| Heat.SI.No :1,2,3,4<br>MP.No : MP19474,MP19475,MP19476,MP19477 , |
|------------------------------------------------------------------|

|                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Remarks:</b><br>1. This material conforms to NACE MR0175 - 2021 / ISO 15156 - 2 - 2020, FTDC001R7 & Material Spec No: RMC 90012 Rev15 & RMP 61158 Rev04.<br>2.Visual inspection as per MSS SP-55-2011 and found accepted. No surface irregularities to Type I none and Type II - XII not exceed 'a'.<br>3.Certified that the Castings are free from Radioactive contaminations.<br>4.MPI carried out as per MSS SP 53 and found accepted. | <div> INSPECTED / REVIEWED &amp;<br/> ACCEPTED / NOT ACCEPTED </div> <div> Sign &amp; Date:  Sample Qty: - </div> <div> M. VADIVEL - FICPL QA </div> |
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We Confirm that the above material was manufactured,sampled,tested & inspected in accordance with the drawing,material specification,Customer order requirement and found to meet the requirements

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|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <br>Prepared By<br><b>SABARI SANKAR</b> | <br>Approved By<br><b>ARUN KUMAR.N</b> |
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|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------------------------------------|
| <br><b>CASTLEGATE</b> | <b>CASTLEGATE INDUSTRIAL HOLDINGS PRIVATE LIMITED</b><br>S.F.No.387/1, Devanampalayam(P.O), Pollachi(TK), Coimbatore, 642 120<br>Phone :7397161500 04259264166, Mail :info@castlegatellc.com |  | <b>ISO 9001 : 2015<br/>CERTIFIED<br/>COMPANY</b> |
|                                                                                                       | <b>INSPECTION CERTIFICATE</b><br>According to EN10204 - 3.1                                                                                                                                  |  |                                                  |
|                                                                                                       | (Empty space for additional details)                                                                                                                                                         |  |                                                  |

|                                                               |                                                                 |
|---------------------------------------------------------------|-----------------------------------------------------------------|
| <b>Customer</b> : FLOWSERVE INDIA CONTROLS PVT LTD<br>CHENNAI | <b>T.C. No</b> : FLS/4027 - 001<br><b>T.C.Date</b> : 13/05/2024 |
| <b>PO.No</b> : 10116580 Dt. 20/02/2024                        | <b>Heat No</b> : 4027<br><b>Pour Date</b> : 13/04/2024          |
| <b>Specification</b> : ASTM A216 GR.WCC (UNS J02503) - 2021   | CASTING PROCESSED BY INDUCTION MELTING<br>FURNACE               |
|                                                               | <b>FOUNDRIY IDENTIFICATION</b> : CG                             |

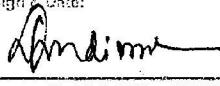
| CHEMICAL COMPOSITION % |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
|------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Element                | C     | Mn    | Si    | P     | S     | Cr    | Ni    | Mo    | Cu    | V     | CE    | RE    | AL    | N     |
| Minimum                | ---   | ---   | ---   | ---   | ---   | ---   | ---   | ---   | ---   | ---   | ---   | ---   | ---   | ---   |
| Maximum                | 0.230 | 1.200 | 0.600 | 0.025 | 0.020 | 0.500 | 0.500 | 0.200 | 0.300 | 0.030 | 0.430 | 1.000 | 0.080 | 0.015 |
| Actual                 | 0.184 | 0.902 | 0.385 | 0.022 | 0.010 | 0.116 | 0.055 | 0.012 | 0.014 | 0.009 | 0.366 | 0.206 | 0.049 | 0.006 |

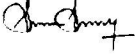
|                         |                                                                                                                                |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>HEAT TREATMENT :</b> | NORMALISING: Temperature raised to 920°C, soaked for 4 Hrs and then atmospheric air cooled.(Cycle No:C0018/24-25Dt.16/04/2024) |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------|

| MECHANICAL PROPERTIES |                |        |                  |        |             |             |          |           |                             |     |     |     |
|-----------------------|----------------|--------|------------------|--------|-------------|-------------|----------|-----------|-----------------------------|-----|-----|-----|
| Requirements          | Yield Strength |        | Tensile Strength |        | %Elongation | % Reduction | Hardness | Bend Test | Impact Test                 |     |     |     |
|                       | kgf/mm²        | Mpa    | kgf/mm²          | Mpa    | GL :50 mm   | In Area     | HBW      |           | CVN 2mm at MIN AVG 27 JULES |     |     |     |
|                       | 0.2 % Rp       |        | -                |        | -           | -           | -        | -         |                             |     |     |     |
| Minimum               | 28.03          | 275    | 49.44            | 485    | 22          | 35          | ---      | ---       |                             |     |     |     |
| Maximum               | ---            | ---    | 66.77            | 655    | ---         | ---         | 235      | ---       | 1                           | 2   | 3   | Avg |
| Actual                | 32.35          | 317.37 | 55.05            | 540.06 | 30.64       | 64.54       | 156      | ---       | ---                         | --- | --- | --- |

| ITEM DESCRIPTION |                     |                 |            |          |
|------------------|---------------------|-----------------|------------|----------|
| SL.No.           | Description         | Drawing No.     | Part No    | Quantity |
| 1                | BODY CSTG 4" CL 900 | 00487876 REV 21 | N487876WCC | 4        |

|                                                                  |
|------------------------------------------------------------------|
| Heat.SI.No :1,2,3,4<br>MP.No : MP19478,MP19479,MP19480,MP19481 , |
|------------------------------------------------------------------|

|                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                    |
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| <b>Remarks:</b><br>1. This material conforms to NACE MR0175 - 2021 / ISO 15156 - 2 - 2020, FTDC001R7 & Material Spec No: RMC 90012 Rev15 & RMP 61158 Rev04.<br>2.Visual inspection as per MSS SP-55-2011 and found accepted. No surface irregularities to Type I none and Type II - XII not exceed 'a'.<br>3.Certified that the Castings are free from Radioactive contaminations.<br>4.MPI carried out as per MSS SP 53 and found accepted. | <div>             INSPECTED / REVIEWED &amp;<br/>             ACCEPTED / NOT ACCEPTED<br/>             Sign &amp; Date:  Sample Qty: -<br/>             M. VADIVEL - FICPL QA           </div> |
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| We Confirm that the above material was manufactured,sampled,tested & inspected in accordance with the drawing,material specification,Customer order requirement and found to meet the requirements |                                                                                                                             |
| <br>Prepared By<br><b>SABARI SANKAR</b>                                                                         | <br>Approved By<br><b>ARUN KUMAR.N</b> |

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|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------------------------------------|
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|                                                                                                       | <b>INSPECTION CERTIFICATE</b><br>According to EN10204 - 3.1                                                                                                                                  |  |                                                  |
|                                                                                                       | (Empty space for additional information)                                                                                                                                                     |  |                                                  |

|                                                               |                                                                 |
|---------------------------------------------------------------|-----------------------------------------------------------------|
| <b>Customer</b> : FLOWSERVE INDIA CONTROLS PVT LTD<br>CHENNAI | <b>T.C. No</b> : FLS/4012 - 001<br><b>T.C.Date</b> : 13/05/2024 |
| <b>PO.No</b> : 10116580 Dt. 20/02/2024                        | <b>Heat No</b> : 4012<br><b>Pour Date</b> : 11/04/2024          |
| <b>Specification</b> : ASTM A216 GR.WCC (UNS J02503) - 2021   | CASTING PROCESSED BY INDUCTION MELTING FURNACE                  |
|                                                               | <b>FOUNDRY IDENTIFICATION</b> : CG                              |

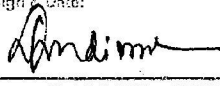
| CHEMICAL COMPOSITION % |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
|------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Element                | C     | Mn    | Si    | P     | S     | Cr    | Ni    | Mo    | Cu    | V     | CE    | RE    | AL    | N     |
| Minimum                | ---   | ---   | ---   | ---   | ---   | ---   | ---   | ---   | ---   | ---   | ---   | ---   | ---   | ---   |
| Maximum                | 0.230 | 1.200 | 0.600 | 0.025 | 0.020 | 0.500 | 0.500 | 0.200 | 0.300 | 0.030 | 0.430 | 1.000 | 0.080 | 0.015 |
| Actual                 | 0.205 | 0.871 | 0.422 | 0.024 | 0.010 | 0.141 | 0.062 | 0.010 | 0.019 | 0.005 | 0.387 | 0.237 | 0.047 | 0.002 |

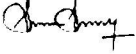
|                         |                                                                                                                                |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>HEAT TREATMENT :</b> | NORMALISING: Temperature raised to 920°C, soaked for 4 Hrs and then atmospheric air cooled.(Cycle No:D0011/24-25Dt.16/04/2024) |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------|

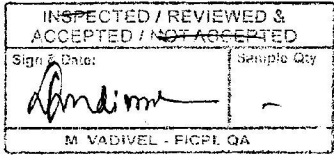
| MECHANICAL PROPERTIES |                |        |                  |        |             |             |          |           |                             |     |     |     |
|-----------------------|----------------|--------|------------------|--------|-------------|-------------|----------|-----------|-----------------------------|-----|-----|-----|
| Requirements          | Yield Strength |        | Tensile Strength |        | %Elongation | % Reduction | Hardness | Bend Test | Impact Test                 |     |     |     |
|                       | kgf/mm²        | Mpa    | kgf/mm²          | Mpa    | GL :50 mm   | In Area     | HBW      |           | CVN 2mm at MIN AVG 27 JULES |     |     |     |
|                       | 0.2 % Rp       |        | -                |        | -           | -           | -        | -         |                             |     |     |     |
| Minimum               | 28.03          | 275    | 49.44            | 485    | 22          | 35          | ---      | ---       |                             |     |     |     |
| Maximum               | ---            | ---    | 66.77            | 655    | ---         | ---         | 235      | ---       | 1                           | 2   | 3   | Avg |
| Actual                | 29.83          | 292.65 | 50.95            | 499.84 | 30.64       | 64.23       | 156      | ---       | ---                         | --- | --- | --- |

| ITEM DESCRIPTION |                     |                 |            |          |
|------------------|---------------------|-----------------|------------|----------|
| SL.No.           | Description         | Drawing No.     | Part No    | Quantity |
| 1                | BODY CSTG 4" CL 900 | 00487876 REV 21 | N487876WCC | 4        |

|                                                                  |
|------------------------------------------------------------------|
| Heat.SI.No :1,2,3,4<br>MP.No : MP19399,MP19400,MP19401,MP19402 , |
|------------------------------------------------------------------|

|                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                    |
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| <b>Remarks:</b><br>1. This material conforms to NACE MR0175 - 2021 / ISO 15156 - 2 - 2020, FTDC001R7 & Material Spec No: RMC 90012 Rev15 & RMP 61158 Rev04.<br>2.Visual inspection as per MSS SP-55-2011 and found accepted. No surface irregularities to Type I none and Type II - XII not exceed 'a'.<br>3.Certified that the Castings are free from Radioactive contaminations.<br>4.MPI carried out as per MSS SP 53 and found accepted. | <div>             INSPECTED / REVIEWED &amp;<br/>             ACCEPTED / NOT ACCEPTED<br/>             Sign &amp; Date:  Sample Qty: -<br/>             M. VADIVEL - FICPL QA           </div> |
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| We Confirm that the above material was manufactured,sampled,tested & inspected in accordance with the drawing,material specification,Customer order requirement and found to meet the requirements |                                                                                                                             |
| <br>Prepared By<br><b>SABARI SANKAR</b>                                                                         | <br>Approved By<br><b>ARUN KUMAR.N</b> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                         |                                                                                                                                                 |                                                                                                                                            |                     |                                                                                                               |                                               |                                   |                  |              |                 |                     |           |            |             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------|------------------|--------------|-----------------|---------------------|-----------|------------|-------------|
|                                                                                                                                                                                                                                                                                                                                                                                              | <b>FLOW LINK SYSTEMS PRIVATE LIMITED</b><br>FOUNDRY UNIT - II<br>65 / 2-A-B & C, PUDUPALAYAM B.P.O<br>AVINASHI TK, COIMBATORE - 641 654<br><b>INDIA</b> |                                                                                                                                                 | <b>INSPECTION CERTIFICATE</b><br>(EN 10204:2004-3.1)                                                                                       |                     | TC NO : <b>A-0833</b><br>DATE : <b>10-May-2024</b>                                                            |                                               |                                   |                  |              |                 |                     |           |            |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>CUSTOMER</b><br><b>FLOWSERVE INDIA CONTROLS PVT LTD,</b><br>B8, MMDA Industrial Area,<br>Maraimalai Nagar-603209<br>Kancheepuram, INDIA              |                                                                                                                                                 | <b>SPECIFICATION REFERENCE</b><br>1. ASTM A216-21 Gr.WCC<br>2. Customer Spec. RMC 90012 Rev. 15<br>3. NACE MR 0175-2021 / ISO 15156 - 2020 |                     | <b>HEAT NO.</b><br><b>0299B</b><br>Date of Pouring<br><b>06-Mar-2024</b><br><b>FOUNDRY MARK</b><br><b>FLS</b> |                                               |                                   |                  |              |                 |                     |           |            |             |
| <b>PO NUMBER &amp; DATE</b><br>10115280 Dt. 05.12.2023                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                         |                                                                                                                                                 | <b>ORDER ACCEPTANCE NO</b><br>24-0223                                                                                                      |                     |                                                                                                               |                                               |                                   |                  |              |                 |                     |           |            |             |
| <b>CHEMICAL COMPOSITION % wt</b>                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                         |                                                                                                                                                 |                                                                                                                                            |                     |                                                                                                               |                                               |                                   |                  |              |                 |                     |           |            |             |
| MELTING PRACTICE : <b>Steel melted using Electric Induction furnace, and fully killed using Aluminium.</b>                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                         |                                                                                                                                                 |                                                                                                                                            |                     |                                                                                                               |                                               |                                   |                  |              |                 |                     |           |            |             |
| <b>SPECIFICATION</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>C</b>                                                                                                                                                | <b>Si</b>                                                                                                                                       | <b>S</b>                                                                                                                                   | <b>P</b>            | <b>Mn</b>                                                                                                     | <b>Ni</b>                                     | <b>Cr</b>                         | <b>Mo</b>        | <b>V</b>     | <b>Cu</b>       | <b>Al</b>           | <b>--</b> | <b>*RE</b> | <b>**CE</b> |
| Minimum                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | --                                                                                                                                                      | --                                                                                                                                              | --                                                                                                                                         | --                  | --                                                                                                            | --                                            | --                                | --               | --           | --              | --                  | --        | --         | --          |
| Maximum                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0.23                                                                                                                                                    | 0.60                                                                                                                                            | 0.020                                                                                                                                      | 0.025               | 1.20                                                                                                          | 0.50                                          | 0.50                              | 0.20             | 0.030        | 0.30            | 0.08                | --        | 1.00       | 0.43        |
| Achieved                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0.189                                                                                                                                                   | 0.414                                                                                                                                           | 0.008                                                                                                                                      | 0.013               | 0.923                                                                                                         | 0.056                                         | 0.109                             | 0.015            | 0.004        | 0.018           | 0.041               | --        | 0.20       | 0.37        |
| <b>Heat Treatment :</b> NORMALIZING: Temperature raised to 920°C, Soaked for 4 hours and then atmospheric air-cooled                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                         |                                                                                                                                                 |                                                                                                                                            |                     |                                                                                                               |                                               |                                   |                  |              |                 |                     |           |            |             |
| <b>Visual Inspection - Castings conforms to ANSI/MSS-SP-55-2011 - Type II to XII (a)</b>                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                         |                                                                                                                                                 |                                                                                                                                            |                     |                                                                                                               |                                               |                                   |                  |              |                 |                     |           |            |             |
| <b>MECHANICAL PROPERTIES</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                         |                                                                                                                                                 |                                                                                                                                            |                     |                                                                                                               |                                               |                                   |                  |              |                 |                     |           |            |             |
| <b>TEST TYPE</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Tensile as per ASTM A370-22 / ISO 6892-1-2020 (at Room Temperature)</b>                                                                              |                                                                                                                                                 |                                                                                                                                            |                     |                                                                                                               | <b>Impact (CVN) ISO 148-2016/ASTM A370-22</b> |                                   | <b>BEND TEST</b> |              | <b>HARDNESS</b> |                     |           |            |             |
| <b>SPECIFICATION</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>YS</b>                                                                                                                                               | <b>YS</b>                                                                                                                                       | <b>UTS</b>                                                                                                                                 | <b>% Elongation</b> |                                                                                                               | <b>% RA</b>                                   | <b>Test Temperature (at °C) J</b> |                  | <b>ANGLE</b> | <b>D</b>        | <b>(HBW / HRBW)</b> |           |            |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Rp 0.2% (KSI)</b>                                                                                                                                    | <b>Rp 1.0% (KSI)</b>                                                                                                                            | <b>(KSI)</b>                                                                                                                               | <b>4d</b>           | <b>5d</b>                                                                                                     |                                               | <b>Values</b>                     | <b>Avg.</b>      |              |                 |                     |           |            |             |
| Minimum                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 40                                                                                                                                                      | --                                                                                                                                              | 70                                                                                                                                         | 22                  | --                                                                                                            | 35                                            | --                                | --               | --           | --              | --                  |           |            |             |
| Maximum                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | --                                                                                                                                                      | --                                                                                                                                              | 95                                                                                                                                         | --                  | --                                                                                                            | --                                            | --                                | --               | --           | --              | 237                 |           |            |             |
| Achieved                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 45.3                                                                                                                                                    | --                                                                                                                                              | 75.1                                                                                                                                       | 34.8                | --                                                                                                            | 64.5                                          | --                                | --               | --           | --              | 152                 |           |            |             |
| <b>DETAIL OF CASTINGS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                         |                                                                                                                                                 |                                                                                                                                            |                     |                                                                                                               |                                               |                                   |                  |              |                 |                     |           |            |             |
| <b>S.No.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>PO S.No.</b>                                                                                                                                         | <b>DESCRIPTION</b>                                                                                                                              | <b>CAST DRAWING NO / REV</b>                                                                                                               | <b>PART NO</b>      | <b>POURED QTY</b>                                                                                             | <b>DESP. QTY</b>                              |                                   |                  |              |                 |                     |           |            |             |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 03                                                                                                                                                      | 4" 900 FE BODY                                                                                                                                  | D-487876 / 20                                                                                                                              | N2757928-60         | 5                                                                                                             | 3                                             |                                   |                  |              |                 |                     |           |            |             |
| <b>REMARKS:</b><br>1. Stated Chemical composition is from heat analysis; and Mechanical properties of test coupon from Cast keel block.<br>2. *RE = Ni + Cr + Mo + V + Cu<br>$** CE = C + \frac{Mn}{6} + \frac{Cr+Mo+V}{5} + \frac{Ni+Cu}{15}$<br>3<br>4. The above stated KSI values have been calculated based on original tested N/mm <sup>2</sup> values.<br>5. Stated actual Hardness value is less than 22 HRC.<br>6. Degassing process conform as per RMP 61496 Rev. A |                                                                                                                                                         |                                                                                                                                                 |                                                                                                                                            |                     |                                                                                                               |                                               |                                   |                  |              |                 |                     |           |            |             |
| <div style="text-align: right;">  </div>                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                         |                                                                                                                                                 |                                                                                                                                            |                     |                                                                                                               |                                               |                                   |                  |              |                 |                     |           |            |             |
| <b>CERTIFIED THAT</b><br>1. THE ABOVE GIVEN DETAILS ARE CORRECT, FULFILLS THE SPECIFICATION / STANDARDS AND PO.<br>2. MATERIAL WAS MANUFACTURED, SAMPLED, TESTED, INSPECTED IN ACCORDANCE WITH THE ABOVE SPECIFICATIONS AND CERTIFIED AS PER EN 10204 - 3.1.<br>3. THE MATERIAL IS FREE OF CONTAMINATION FROM RADIO ACTIVE ELEMENT OR RADIATION.                                                                                                                              |                                                                                                                                                         |                                                                                                                                                 |                                                                                                                                            |                     |                                                                                                               |                                               |                                   |                  |              |                 |                     |           |            |             |
| PREPARED BY:<br><br><b>S.KARTHICK</b>                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                         | CHECKED BY:<br><br><b>A.LOGANATHAN</b><br><b>LAB CHEMIST</b> |                                                                                                                                            |                     |                                                                                                               |                                               |                                   |                  |              |                 |                     |           |            |             |
| <div style="text-align: right;"> <br/> <b>M.S.DHANABALAN</b><br/> <b>DEPUTY MANAGER - QUALITY</b><br/> <b>MANUFACTURER'S AUTHORIZED INSPECTOR</b> </div>                                                                                                                                                                                                                                 |                                                                                                                                                         |                                                                                                                                                 |                                                                                                                                            |                     |                                                                                                               |                                               |                                   |                  |              |                 |                     |           |            |             |

|                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                              |        |                         |        |                    |                    |                                        |                  |                             |                                                                                                                                                |                                                  |       |       |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------------------|--------|--------------------|--------------------|----------------------------------------|------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-------|-------|-------|
| <br><b>CASTLEGATE</b>                                                                                                                                                                                                                                                                                                                                                                                                          | <b>CASTLEGATE INDUSTRIAL HOLDINGS PRIVATE LIMITED</b><br>S.F.No.387/1, Devanampalayam(P.O), Pollachi(TK), Coimbatore, 642 120<br>Phone :7397161500 04259264166, Mail :info@castlegatellc.com |        |                         |        |                    |                    |                                        |                  |                             |                                                                                                                                                | <b>ISO 9001 : 2015<br/>CERTIFIED<br/>COMPANY</b> |       |       |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>INSPECTION CERTIFICATE</b><br>According to EN10204 - 3.1                                                                                                                                  |        |                         |        |                    |                    |                                        |                  |                             |                                                                                                                                                |                                                  |       |       |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                              |        |                         |        |                    |                    |                                        |                  |                             |                                                                                                                                                |                                                  |       |       |       |
| <b>Customer</b> : FLOWSERVE INDIA CONTROLS PVT LTD<br>CHENNAI                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                              |        |                         |        |                    |                    |                                        |                  |                             | <b>T.C. No</b> : FLS/3977 - 001<br><b>T.C.Date</b> : 25/04/2024                                                                                |                                                  |       |       |       |
| <b>PO.No</b> : 10116580 Dt. 20/02/2024<br><br><b>Specification</b> : ASTM A216 GR.WCC (UNS J02503) - 2021                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                              |        |                         |        |                    |                    |                                        |                  |                             | <b>Heat No</b> : 3977<br><b>Pour Date</b> : 09/04/2024<br>CASTING PROCESSED BY INDUCTION MELTING FURNACE<br>FOUNDRY IDENTIFICATION : <b>CG</b> |                                                  |       |       |       |
| <b>CHEMICAL COMPOSITION %</b>                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                              |        |                         |        |                    |                    |                                        |                  |                             |                                                                                                                                                |                                                  |       |       |       |
| <b>Element</b>                                                                                                                                                                                                                                                                                                                                                                                                                 | C                                                                                                                                                                                            | Mn     | Si                      | P      | S                  | Cr                 | Ni                                     | Mo               | Cu                          | V                                                                                                                                              | CE                                               | RE    | AL    | N     |
| <b>Minimum</b>                                                                                                                                                                                                                                                                                                                                                                                                                 | ---                                                                                                                                                                                          | ---    | ---                     | ---    | ---                | ---                | ---                                    | ---              | ---                         | ---                                                                                                                                            | ---                                              | ---   | ---   | ---   |
| <b>Maximum</b>                                                                                                                                                                                                                                                                                                                                                                                                                 | 0.230                                                                                                                                                                                        | 1.200  | 0.600                   | 0.025  | 0.020              | 0.500              | 0.500                                  | 0.200            | 0.300                       | 0.030                                                                                                                                          | 0.430                                            | 1.000 | 0.080 | 0.015 |
| <b>Actual</b>                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.210                                                                                                                                                                                        | 0.952  | 0.426                   | 0.024  | 0.009              | 0.229              | 0.072                                  | 0.020            | 0.017                       | 0.006                                                                                                                                          | 0.426                                            | 0.344 | 0.050 | 0.007 |
| <b>HEAT TREATMENT :</b> NORMALISING: Temperature raised to 920°C, soaked for 4 Hrs and then atmospheric air cooled.(Cycle No:D0009/24-25Dt.12/04/2024)                                                                                                                                                                                                                                                                         |                                                                                                                                                                                              |        |                         |        |                    |                    |                                        |                  |                             |                                                                                                                                                |                                                  |       |       |       |
| <b>MECHANICAL PROPERTIES</b>                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                              |        |                         |        |                    |                    |                                        |                  |                             |                                                                                                                                                |                                                  |       |       |       |
| Requirements                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Yield Strength</b>                                                                                                                                                                        |        | <b>Tensile Strength</b> |        | <b>%Elongation</b> | <b>% Reduction</b> | <b>Hardness</b>                        | <b>Bend Test</b> | <b>Impact Test</b>          |                                                                                                                                                |                                                  |       |       |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                | kgf/mm <sup>2</sup>                                                                                                                                                                          | Mpa    | kgf/mm <sup>2</sup>     | Mpa    | GL :50 mm          | In Area            | HBW                                    |                  | CVN 2mm at MIN AVG 27 JULES |                                                                                                                                                |                                                  |       |       |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.2 % Rp                                                                                                                                                                                     |        | -                       |        | -                  | -                  | -                                      | -                |                             |                                                                                                                                                |                                                  |       |       |       |
| <b>Minimum</b>                                                                                                                                                                                                                                                                                                                                                                                                                 | 28.03                                                                                                                                                                                        | 275    | 49.44                   | 485    | 22                 | 35                 | ---                                    | ---              |                             |                                                                                                                                                |                                                  |       |       |       |
| <b>Maximum</b>                                                                                                                                                                                                                                                                                                                                                                                                                 | ---                                                                                                                                                                                          | ---    | 66.77                   | 655    | ---                | ---                | 237                                    | ---              | 1                           | 2                                                                                                                                              | 3                                                | Avg   |       |       |
| <b>Actual</b>                                                                                                                                                                                                                                                                                                                                                                                                                  | 30.12                                                                                                                                                                                        | 295.43 | 50.62                   | 496.56 | 30.72              | 60.38              | 163                                    | ---              | ---                         | ---                                                                                                                                            | ---                                              | ---   |       |       |
| <b>ITEM DESCRIPTION</b>                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                              |        |                         |        |                    |                    |                                        |                  |                             |                                                                                                                                                |                                                  |       |       |       |
| SL.No.                                                                                                                                                                                                                                                                                                                                                                                                                         | Description                                                                                                                                                                                  |        |                         |        |                    | Drawing No.        |                                        | Part No          |                             |                                                                                                                                                | Quantity                                         |       |       |       |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                              | BODY CSTG 4" CL 900                                                                                                                                                                          |        |                         |        |                    | 00487876 REV 21    |                                        | N487876WCC       |                             |                                                                                                                                                | 4                                                |       |       |       |
| Heat.SI.No :1,2,3,4<br>MP.No : MP19274,MP19275,MP19276,MP19277 ,                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                              |        |                         |        |                    |                    |                                        |                  |                             |                                                                                                                                                |                                                  |       |       |       |
| Remarks:<br>1. This material conforms to NACE MR0175 - 2021 / ISO 15156 - 2 - 2020, FTDC001R7 & Material Spec No: RMC 90012 Rev15.<br>2.Visual inspection as per MSS SP-55-2011 and found accepted. No surface irregularities to Type I none and Type II - XII not exceed 'a'.<br>3.Certified that the Castings are free from Radioactive contaminations.<br>4.MPI carried out as per ASME SECTION V Art.7 and found accepted. |                                                                                                                                                                                              |        |                         |        |                    |                    |                                        |                  |                             |                                                                                                                                                |                                                  |       |       |       |
| <div style="border: 1px solid black; padding: 5px; display: inline-block;">             INSPECTED / REVIEWED &amp;<br/>             ACCEPTED / NOT ACCEPTED<br/>             Sign &amp; Date:  Sample Qty: -<br/>             M. VADIVEL - FICPL QA           </div>                                                                                                                                                           |                                                                                                                                                                                              |        |                         |        |                    |                    |                                        |                  |                             |                                                                                                                                                |                                                  |       |       |       |
| We Confirm that the above material was manufactured,sampled,tested & inspected in accordance with the drawing,material specification,Customer order requirement and found to meet the requirements                                                                                                                                                                                                                             |                                                                                                                                                                                              |        |                         |        |                    |                    |                                        |                  |                             |                                                                                                                                                |                                                  |       |       |       |
| <br>Prepared By<br><b>SABARI SANKAR</b>                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                              |        |                         |        |                    |                    | <br>Approved By<br><b>ARUN KUMAR.N</b> |                  |                             |                                                                                                                                                |                                                  |       |       |       |